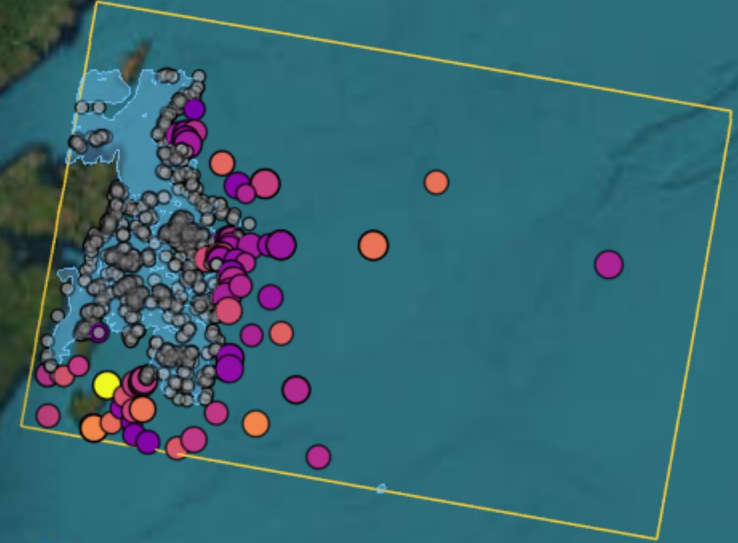


PROJECT BRIEF · SENTINEL-1 SAR

Iceberg Alley SAR

Detecting icebergs off Newfoundland in satellite radar, and showing them on an interactive globe with a timeline.

Natali Kovalski · October 2026



UTC May 7 2025 00:00:00 UTC May 9 2025 00:00:00 UTC May 11 2025 00:00:00 UTC May 13 2025 00:00:00 UTC May 15 2025 00:00:00 UTC

THE PROBLEM

Every spring, icebergs drift into busy water

01

The hazard

Greenland bergs ride the Labrador Current down "Iceberg Alley", past shipping lanes, fishing grounds and offshore platforms.

02

The gap

Aircraft reconnaissance is costly and limited by weather, and this coast is often fogged in.

03

The idea

Satellite radar sees through cloud and darkness. Sentinel-1 data is free and revisits every few days.

APPROACH

Six steps from raw scene to globe

01

Search

ASF search for HH+HV scenes over the AOI

02

Calibrate

HyP3 terrain correction: gamma0, linear power, 10 m

03

Mask

Land plus 500 m from OSM; pack ice from HV texture

04

Detect

CA-CFAR on HV, threshold set from the scene's speckle

05

Filter

Size limits, ice-strip rejection, HH co-pol check

06

Publish

GeoJSON with attributes, then CZML for the Cesium viewer

Every parameter lives in **config.yaml**, with lengths in metres so tuning carries over between 10 m and 20 m pixels.

Four ideas behind the detector

Backscatter

Calm water reflects radar away and looks dark. An iceberg's facets send energy back and look bright.

Polarization

Sea clutter is far weaker in cross-pol (HV), so bergs stand out there. HH is the second check.

Speckle

Grainy noise in every SAR image. Its strength (ENL) is measured per scene and sets the threshold.

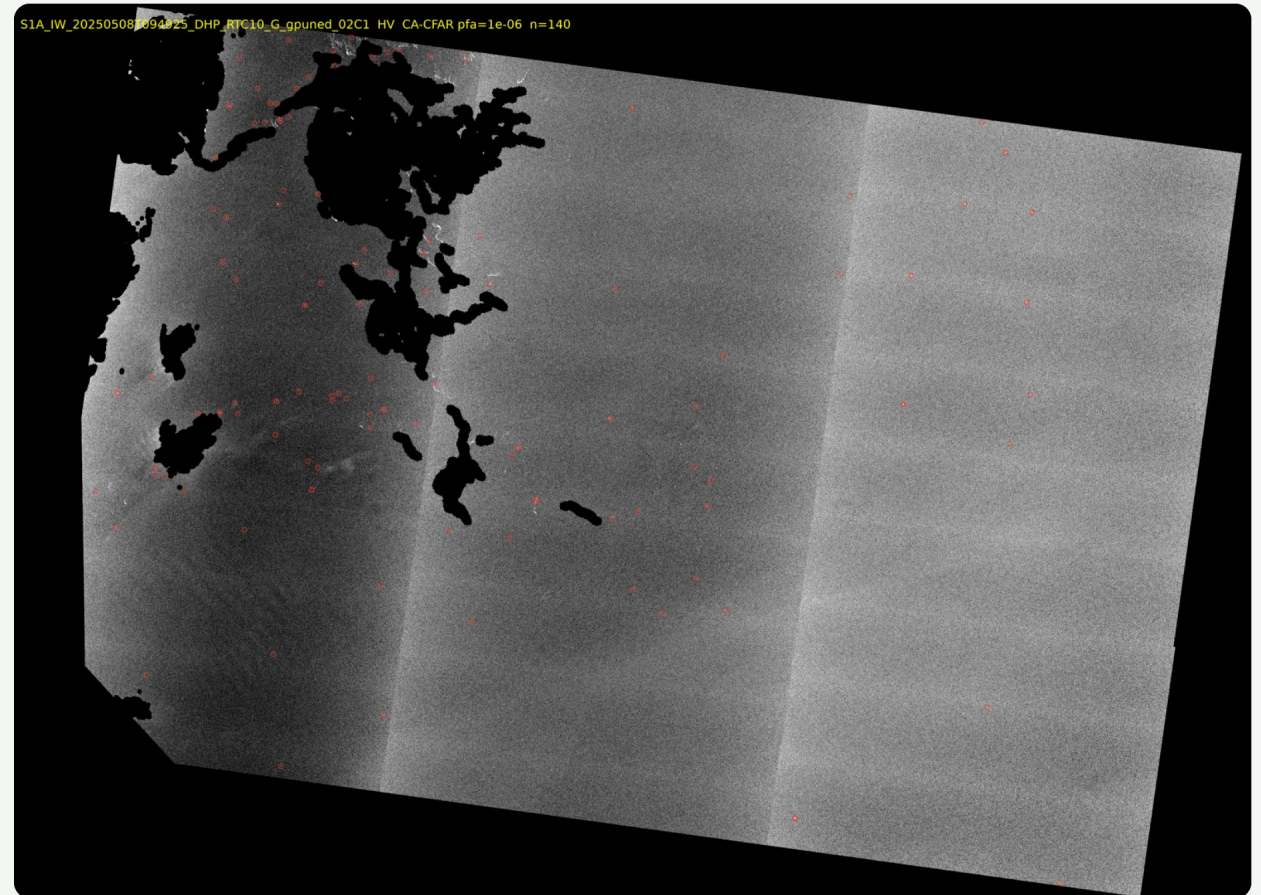
CFAR

Each pixel is compared with its local background, so false alarms stay rare in calm and rough seas.

DETECTION

CFAR on HV, then three filters

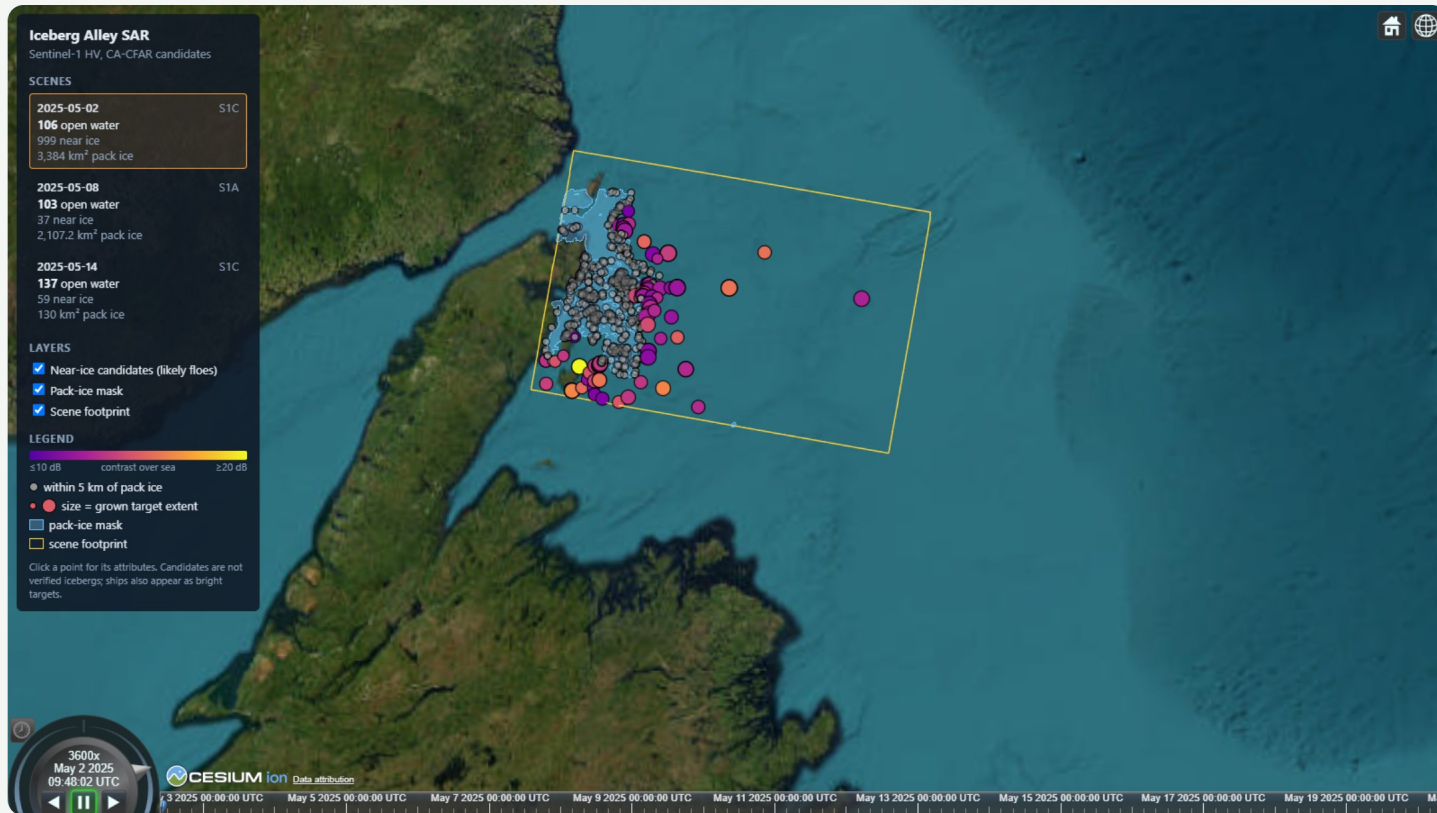
- **Threshold** from a 10^{-6} false-alarm rate, speckle measured per scene
- **Size:** 2 px to 0.8 km²
- **Ice strips:** reject bright structures longer than 300 m
- **HH check:** at least 6 dB above background in both polarizations



HV, 2025-05-08. Land and pack ice masked (black), 140 detections circled.

VIEWER

Every candidate on a globe, by date



- Scrub the timeline through the spring 2025 passes
- Click a target to see its attributes
- Colour shows contrast, size shows extent
- Grey: within 5 km of pack ice, likely floes

HEADLINE RESULT

78%

of the icebergs that International Ice Patrol analysts marked on the same Sentinel-1 pass were found: **209 of 268**.

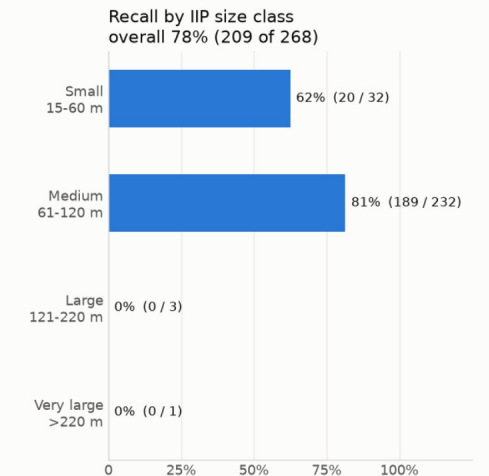
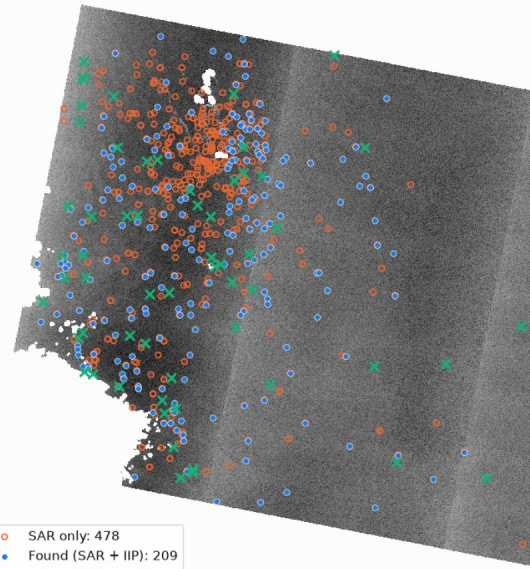
2019-04-29 · median position difference 128 m · recall steady at 73–84% from the coast to 100 km offshore

RECALL BY SIZE

Medium bergs are found best

Berg size	Recall
Small, 15–60 m	62%
Medium, 61–120 m	81%
Large, over 120 m	0 of 4
All	78%

Detections vs. IIP analyst labels, same Sentinel-1 pass



2019-04-29 09:41 UTC, Sentinel-1A IW, HV, 10 m.
IIP = International Ice Patrol sightings DB (NSIDC G00807).
Median match distance 128 m.

Detections vs. IIP analyst labels, 2019-04-29.

VALIDATION

Every ground-truth source has a catch

Source	What it gives	Outcome
NAIS iceberg chart	Counts per 1° square	Within 2–4×, but opposite errors cancel
IIP aircraft sightings	Positions 5–7 h after the pass	Inconclusive: drift unresolved
Sentinel-2 optical	Bright objects at 10 m	Loose sea ice in April, bergs inshore in June
IIP labels, same pass	Positions on the same image	Headline: 78% recall, no drift

Where truth comes hours later, the code estimates a common drift offset and checks it against chance. If the peak is weak, it reports "drift unresolved" instead of a score.

What the numbers don't show yet

Limits

- **Precision unknown:** 475 of 684 detections are unlabelled bergs, boats or false alarms
- **Large bergs missed:** 0 of 4 over 120 m
- **Rough seas** raise the HV background 3–4 dB
- **Near pack ice**, many targets are floes (flagged, not removed)

Next

- **AIS ship positions** for the same passes, to measure precision
- **Size-aware filters:** scale the strip filter and guard window to the target
- **OS-CFAR or K-distribution** for heavy-tailed sea clutter
- **Docker** and a hosted static viewer

BUILD

One command, end to end

```
python -m iceberg_sar.cli run --start 2025-05-01 --end 2025-05-15
```

Processing

Python 3.11, numpy, scipy,
rasterio, GeoPandas, typer CLI

Data

asf_search, HyP3 RTC, OSM
land polygons, IIP sightings

Viewer

Vue 3, Vite, CesiumJS, time-
dynamic CZML

Quality

67 pytest tests, all parameters
in config.yaml

Re-runs reuse downloaded scenes and existing outputs. A dry run shows the plan and the HyP3 credit cost before anything is ordered.

THANK YOU

Questions?

Natali Kovalski · github.com/natali-kovalski/Icebergs

Contains modified Copernicus Sentinel data (2019–2025). Iceberg data: International Ice Patrol (NSIDC G00807) and the North American Ice Service. Land polygons © OpenStreetMap contributors.